CYB495 – Information and Cybersecurity Capstone

Final Project Journal

5.2 Assignment: Journal Final (IE Assessment)

Healthcare ISP Network Infrastructure and Security Architecture

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Network Architect/Engineer, Team Echo
Team members: Ernest Foubi and three teammates
Course instructor feedback incorporated throughout
ECPI University
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Project Overview

Team Echo responded to a Request for Proposal (RFP) for a healthcare Internet Service Provider (ISP) that securely connects the National Institutes of Health (NIH), Cedars-Sinai Medical Center, and Mayo Clinic to the Internet and to a centralized electronic health record (EHR) repository. The RFP required secure Internet and EHR access, redundant connectivity, 24/7 network operations, and scalable, compliant infrastructure.

As the team's Network Architect/Engineer, I was responsible for WAN connectivity, VLANs, IP addressing, routing, network segmentation, redundancy, and network topology. This journal combines my five weekly entries, followed by a final reflection that evaluates the project as a whole. Each weekly entry follows the same structure: contributions, obstacles, visual evidence, resources, legal and ethical considerations, security considerations, and evaluation.

Summary of My Contributions by Week

Week	Project phase	My main contributions
1	Planning and requirements	Accepted the Network Architect role; wrote the Network Requirements section of the Detailed Requirements Report; drafted the initial logical network architecture
2	Network mock-up	Built the Cisco Packet Tracer mock-up (WAN /30 links, VLAN 10/30, router-on-a-stick, routing); verified Guest isolation; tested NIH OSPF/HSRP failover
3	RFP presentation	Revised WAN redundancy after instructor feedback (diverse paths, 5G backup); supplied failover evidence; reviewed the master Packet Tracer configurations
4	Physical implementation	Established the greenfield assumption and equipment quantities; reviewed WAN design; corrected the annual recurring cost
5	Final presentation	Presented slides 16–19 (disaster recovery, budgets, closing recommendation); verified the final cost figures
5+	Corrections and validation	Applied and tested the Week 3 corrections in my own copy of the Packet Tracer file: OSPF at all sites, ASA0 failover path, tighter Guest ACLs, backup DHCP

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1.3 Assignment: Week 1 Journal

Student	Ernest Foubi
Team	Team Echo
Role	Network Architect/Engineer
Week	Week 1 – Project Planning and Requirements Analysis

1. Weekly Contributions

During Week 1, I reviewed the healthcare ISP Request for Proposal (RFP) and joined team discussions to define project responsibilities. I accepted the **Network Architect/Engineer role**, focusing on the following areas:

- WAN connectivity
- VLANs and IP addressing
- Routing and network segmentation
- Redundancy and network topology

I created the Network Requirements section for our team's Detailed Requirements Report. My input detailed how the proposed ISP would connect the National Institutes of Health, Cedars-Sinai, and Mayo Clinic securely, ensuring dependable Internet access, protected electronic health records (EHRs), and room for future growth. Additionally, I provided an initial logical network architecture diagram and suggested using Cisco Packet Tracer to verify the network design in Week 2.

2. Challenges and Solutions

A key challenge was defining clear responsibilities, as some team members initially showed interest in both cybersecurity and network design tasks. Through our discussions, we developed a proposed division of responsibilities across areas such as network architecture, cybersecurity/compliance, cloud infrastructure/storage, and project management.

Another challenge was determining how to separate regular Internet traffic from sensitive patient information if a healthcare organization's credentials or VPN connection became compromised. I proposed the following controls:

- VLAN segmentation
- Firewalls
- Access Control Lists (ACLs)
- Encrypted connections
- Restricted access to protected EHR systems

3. Documentation and Project Progress

I contributed the Network Requirements section and an initial network architecture concept to the shared report. The design identifies the three healthcare locations, secure WAN connectivity, the ISP security perimeter, and the centralized cloud/data center environment.

The team also used Microsoft Teams to coordinate responsibilities, share documents, discuss technical requirements, and review project progress. We developed our Week 1 report and are awaiting final team review before submission.

4. Sources and Resources

The primary resources used for my work were:

- CYB495 Healthcare ISP Request for Proposal (RFP)
- Course project instructions
- Team Echo shared Detailed Requirements Report

These resources helped identify requirements for network connectivity, redundancy, scalability, security, and availability.

5. Legal, Ethical, and Security Considerations

Since the proposed ISP will manage sensitive patient data, I focused on HIPAA security standards, confidentiality, secure communication, and access control. My network design prioritizes segmentation, encryption, restricted access, redundancy, and continuous monitoring to ensure the confidentiality, integrity, and availability of healthcare information.

6. Progress Evaluation and Next Steps

I believe Week 1 established a foundation for the project through team coordination, defined responsibilities, and the development of our initial network requirements. My main contribution was translating the RFP into practical network design requirements.

During Week 2, I plan to work with my teammates to:

- Develop the network topology in Cisco Packet Tracer
- Configure VLANs and routing
- Establish connectivity between the healthcare locations
- Test network segmentation and redundancy where supported

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2.2 Assignment: Week 2 Journal

Student	Ernest Foubi
Team	Team Echo
Project role	Network design, WAN/VLAN architecture, routing, and connectivity verification
Week	Week 2 – Network Mock-up

1. Weekly Contributions

This week, I advanced Team Echo's simulated healthcare network by developing and validating the underlying connectivity between the National Institutes of Health (NIH), Cedars-Sinai Medical Center, Mayo Clinic, and the centralized ISP-hosted Electronic Health Records (EHR) and DNS servers. My primary responsibility was network architecture and technical verification, while other team members led firewall/Internet services, security policy, and server/storage work.

1.1 Initial Network Mock-up

- **Built the network mock-up:** Built the initial Cisco Packet Tracer network mock-up using Cisco 2911 routers, 2960 switches, three hospital sites, Clinical and Guest PCs, and centralized EHR/DNS servers. Established three point-to-point WAN /30 networks (172.16.0.0/30, 172.16.0.4/30, and 172.16.0.8/30).
- **Configured VLANs and routing:** Configured and verified VLAN 10 (Clinical) and VLAN 30 (Guest), 802.1Q router-on-a-stick gateways, and static routing. Hospital Clinical networks use 10.20.10.0/24, 10.30.10.0/24, and 10.40.10.0/24; Guest networks use the corresponding 10.x.30.0/24 ranges.
- **Reviewed the expanded topology:** Reviewed the project lead's expanded topology with an ASA firewall, simulated Internet router, external server, DHCP, and DNS. Independently checked WAN interfaces, default/static routes, server addressing, and the six PCs' DHCP leases.
- **Verified Guest isolation:** Verified the security lead's updated Guest-isolation configuration across all three hospitals: all six PCs passed planned addressing and connectivity checks; the three ACLs were applied inbound on each router's GigabitEthernet0/0.30 subinterface; and all nine web-access tests produced the expected results.

1.2 Redundancy and Dynamic Routing (NIH)

After the project lead expanded the network with secondary hospital routers, a second ISP core, additional backend switching, and a secondary firewall, I investigated the NIH recovery path in a separate Packet Tracer working copy. The original design used static routes; testing showed that a static route could remain selected even when the primary hospital LAN was unavailable. I helped transition the NIH routes to OSPF so the core could react to route withdrawal rather than depend on manual route edits.

- **OSPF adjacency:** Confirmed OSPF adjacency between NIH-R2 and ISP-CORE-R2, between NIH-R1 and ISP-CORE-R1, and across the ISP core-to-core transit link (172.16.0.12/30). Verified FULL neighbor states with show ip ospf neighbor.
- **Backup route selection:** Checked route-selection evidence with show ip route: while NIH-R1's LAN was down, ISP-CORE-R1 learned both NIH Clinical 10.20.10.0/24 and Guest 10.20.30.0/24 through OSPF via ISP-CORE-R2 (172.16.0.14; metric 66).
- **HSRP preemption:** Restored NIH-R1's LAN and confirmed HSRP preemption: NIH-R1 became Active for Clinical group 10 and Guest group 30; NIH-R2 returned to Standby. OSPF preferred the primary NIH path via 172.16.0.2 (metric 65).
- **Automatic failover:** Simulated a second NIH-R1 LAN outage without manually changing routes. HSRP automatically made NIH-R2 Active for both VLANs, and ISP-CORE-R1 automatically selected the backup OSPF path via 172.16.0.14. Restoring the LAN returned HSRP and OSPF to their normal primary paths.

1.3 Team Coordination

The project lead reported dual hospital WAN/router connections, ISP core and switch redundancy, backup ASA deployment, and failure tests. The cloud lead reported secondary DNS at 10.10.10.21, standby EHR at 10.10.10.11, and backup server at 10.10.10.30, along with preliminary service and failover checks. These team-reported additions are distinct from my independently verified NIH OSPF/HSRP and endpoint tests. Backup-file restoration testing remained pending in the cloud lead's update.

1.4 Addressing Plan Summary

Network	VLAN / Type	Address ranges
WAN links (ISP core to hospitals)	Point-to-point /30	172.16.0.0/30, 172.16.0.4/30, 172.16.0.8/30
ISP core-to-core transit link	Point-to-point /30	172.16.0.12/30
Hospital Clinical networks	VLAN 10	10.20.10.0/24, 10.30.10.0/24, 10.40.10.0/24
Hospital Guest networks	VLAN 30	Corresponding 10.x.30.0/24 ranges
EHR servers	Primary / standby	10.10.10.10 / 10.10.10.11
DNS servers	Primary / secondary	10.10.10.20 / 10.10.10.21
Backup server	Server	10.10.10.30

2. Network Mock-up and Visual Evidence

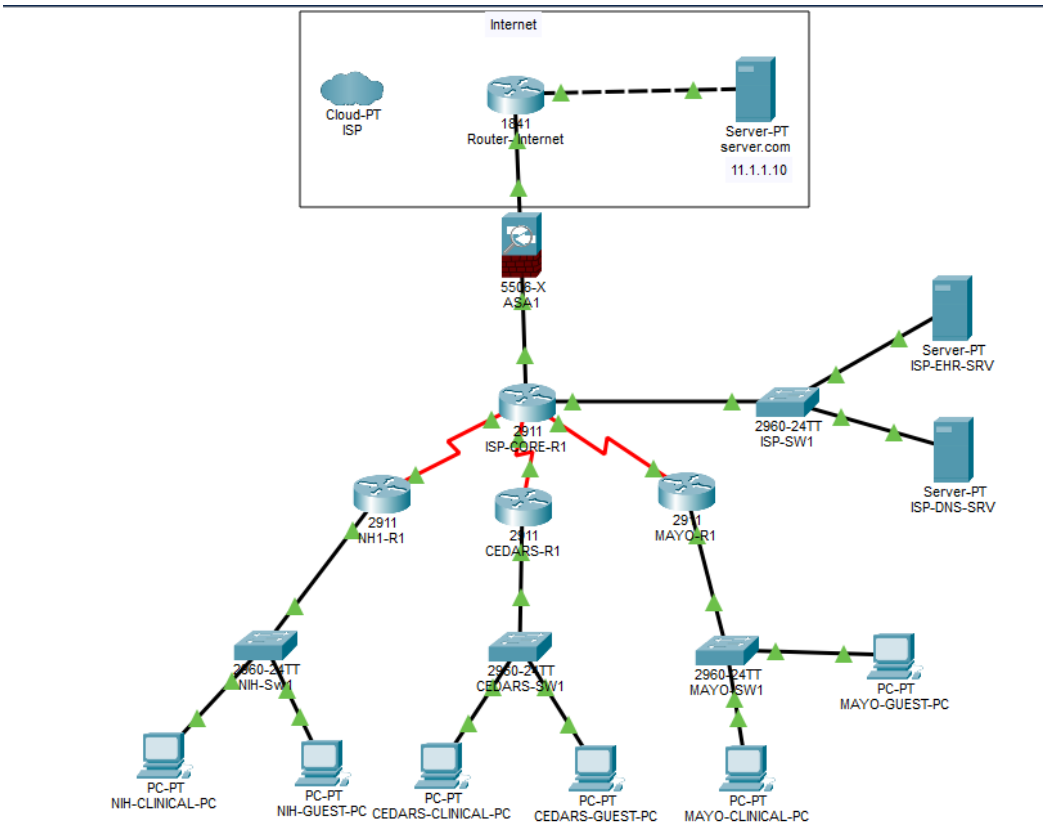


Figure 1. Initial Week 2 Packet Tracer topology before the later redundant-router, dual-core, dual-firewall, and expanded server design. The team's updated topology should be included in the cumulative final journal.

Supporting verification evidence collected during testing includes:

- Command-prompt screenshots (***ipconfig /all and ping***)
- Router ***show ip access-lists and running-configuration output***
- Browser screenshots showing authorized EHR access, Guest timeouts, and successful external web access

These screenshots can be appended to the cumulative final journal.

3. Obstacles and How I Addressed Them

Obstacle	What happened	How I addressed it
Version incompatibility	The security lead's updated .pkt file required Packet Tracer 9.0.1.0858, while my installed version was 9.0.0.0810.	I identified the version mismatch and used the newer compatible software before continuing verification.
Security gaps in an earlier file	NIH Guest initially reached the Clinical PC and EHR server.	Rather than independently modifying a teammate's master configuration, I reported the finding, coordinated with the security lead, and waited for the updated security file.

Obstacle	What happened	How I addressed it
DNS policy inconsistency	An earlier DHCP review found Cedars and Mayo Guest clients had no DNS server assigned.	The security lead subsequently updated DHCP/DNS and ACL rules; my later tests confirmed that all three Guest PCs received DNS server 10.10.10.20 and resolved server.com.
Intermittent first-ping timeout	Several Clinical PCs initially lost one packet to the external server, but subsequent four-packet tests succeeded.	I repeated the tests rather than treating a single timeout as a persistent failure.
Static routes did not fail over	Testing showed that a static route could remain selected even when the primary hospital LAN was unavailable.	I helped transition the NIH routes to OSPF so the core could react to route withdrawal rather than depend on manual route edits (Section 1.2).
Secondary firewall (ASA0) return traffic	NAT translation hits observed, but return Internet traffic through ASA0 not validated.	Investigated ASA0 interfaces, default route, and NAT (details below); recorded as an open issue rather than represented as completed.

The ASA0 investigation showed that its inside/outside interfaces and default route were present, ASA0 could ping the simulated Internet router and server, and NAT translations were recorded. These facts alone did not prove end-to-end client access through the backup firewall. Return traffic or routing/NAT symmetry remains an unresolved concern; I did not represent secondary firewall failover as completed.

4. Technical Resources and Their Use

The principal resources were:

- Cisco Packet Tracer device interfaces and topology
- Router CLI outputs (***show ip interface brief, show ip route, show ip access-lists, show running-config, show ip protocols, show ip ospf neighbor, show ip ospf interface, show standby brief***)
- ASA command output (***show nat, show xlate***)
- PC ipconfig, ping, and tracert output
- The team's shared topology, addressing plan, and implementation plans
- Teammate updates in Microsoft Teams, including the project and cloud leads' reports
- Cisco Networking Academy's Packet Tracer download resource, relevant to resolving the version mismatch: https://www.netacad.com/resources/lab-downloads
- The course assignment instructions and rubric, which guided the journal's evidence and reflection sections

I evaluated these resources by comparing configured addresses and ACL entries with live test results rather than relying solely on reported success. I also compared routing-table next hops and HSRP states against actual endpoint test results. This provided stronger evidence than topology appearance or configuration alone.

5. Legal, Ethical, and Security Considerations

5.1 Legal and Ethical

Because the project represents a healthcare environment, access to EHR resources must follow least-privilege principles and protect patient information confidentiality. HIPAA provides an important regulatory context for safeguarding electronic protected health information, but a successful Packet Tracer simulation does not establish HIPAA compliance. Our tests used simulated systems rather than real patient records; neither successful pings nor firewall ACLs establish HIPAA compliance.

5.2 Security Controls Verified

I confirmed that Clinical PCs could reach the EHR portal while Guest PCs could not. Each Guest ACL:

- First permits DNS requests to 10.10.10.20 over UDP/TCP port 53
- Then denies the three Clinical subnets and the ISP server subnet
- Is applied inbound on VLAN 30

5.3 Security During Failover

Healthcare availability is important, but failover must not bypass access restrictions. During NIH backup operation, Guest access to the tested EHR and Clinical addresses remained blocked while external name resolution and connectivity worked. This supports least privilege and separation of guest traffic from clinical resources in the tested scenarios.

5.4 Remaining Security Concerns

A remaining design concern is the final permit ip any any rule: other internal destinations, including infrastructure or other Guest networks, may remain accessible. I flagged this for the security and project leads rather than changing another team member’s configuration without approval. Further review should address Guest-to-Guest isolation, management-plane protection, AAA/SSH, secure authentication, logging, monitoring, resilience, and both firewall paths.

6. Evaluation of Progress and Quality

6.1 Initial Mock-up Verification

The mock-up is operational for the tested scenarios. All six hospital PCs had valid DHCP addressing and DNS settings; all three Guest PCs were blocked from the tested Clinical and EHR destinations while retaining external connectivity; and all three Clinical PCs reached the EHR server. Application testing yielded nine expected outcomes: each Clinical PC opened the EHR webpage, each Guest PC timed out when opening the EHR webpage, and each Guest PC opened the

simulated external website by hostname. Router inspection also confirmed the three named Guest ACLs and their inbound VLAN 30 placement.

Test	Devices	Result
DHCP addressing and DNS settings	6 hospital PCs	Valid addressing and DNS settings
Clinical PC → EHR webpage	3 Clinical PCs	Opened (expected)
Guest PC → EHR webpage	3 Guest PCs	Timed out (expected – blocked)
Guest PC → external website by hostname	3 Guest PCs	Opened (expected)
Guest ACL placement	3 hospital routers	Named ACLs applied inbound on VLAN 30

6.2 NIH Failover Test Results

Test / checkpoint	Observed evidence	Assessment
NIH HSRP takeover	NIH-R2 Active for groups 10 and 30 during NIH-R1 LAN shutdown	Pass
OSPF backup route	Both NIH /24 routes via 172.16.0.14, OSPF metric 66	Pass
Clinical EHR connectivity	10.10.10.10: 4/4 ICMP replies during failover	Pass — network reachability
Clinical DNS reachability	10.10.10.20 and 10.10.10.21: 4/4 replies each	Pass — ICMP only
Clinical Internet/DNS	11.1.1.10 and server.com: 4/4 replies each	Pass — ping and name resolution
Guest EHR/Clinical restrictions	EHR and NIH Clinical PC: 0/4, unreachable from Guest gateway	Pass for tested destinations
Guest Internet/DNS	server.com resolved; 3/4 replies (first ping timed out)	Pass with initial loss
Primary restoration	NIH-R1 Active, NIH-R2 Standby; OSPF routes via 172.16.0.2, metric 65	Pass
Secondary firewall ASA0	NAT translation hits observed, but return Internet traffic through ASA0 not validated	Open issue

6.3 Overall Evaluation

These initial results demonstrated basic routing, segmentation, DNS, and HTTP connectivity. My original WAN/VLAN network mock-up then progressed to a more resilient multi-router design. The strongest verified Week 2 result is NIH's automatic HSRP gateway takeover, OSPF backup-route selection, and restoration to primary operation, with Clinical connectivity and tested Guest isolation maintained during failure.

Limits include the unresolved ASA0 return path, incomplete end-to-end validation of the updated master file, and the fact that ICMP tests alone do not verify application availability or data integrity. This remains a simulated design, not a production-ready or HIPAA-certified network. Remaining tasks:

- Review and strengthen Guest ACL coverage, including management and inter-Guest restrictions.
- Integrate and retest the cloud lead's secondary DNS, standby EHR, and backup server changes in the consolidated file.
- Troubleshoot secondary ASA0 Internet return traffic; test actual firewall-path failover.

- Retain failover evidence, backup/recovery documentation, and an agreed team checkpoint.
- Complete consolidated testing across NIH, Cedars-Sinai, and Mayo Clinic.

7. Next Steps

- Share verified NIH OSPF/HSRP failover findings with the project lead, the security lead, and the cloud lead and coordinate integration into the team master Packet Tracer file.
- Resolve the outstanding ASA0 return-path/NAT issue, then retest EHR, DNS, Internet access, and Guest restrictions in the consolidated topology.
- Preserve a separate tested working copy, collect dated CLI/browser screenshots, and append updated topology evidence to the cumulative final journal.

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3.2 Assignment: Week 3 Journal

Student	Ernest Foubi
Team	Team Echo (Ernest Foubi and three teammates)
Project role	Network Architect/Engineer – WAN architecture, redundancy, routing, and failover
Week	Week 3 – RFP Presentation

1. Weekly Contributions

This week, I focused on supporting Team Echo's RFP presentation by reviewing our proposed healthcare ISP network architecture and ensuring that our technical recommendations align with the project requirements.

As Network Architect/Engineer, I focused on the network architecture, WAN redundancy, and failover content, and on making sure the presentation accurately separated what our mock-up demonstrated from what we are proposing for production.

- **Revised WAN redundancy after instructor feedback:** Our Week 2 design relied mainly on redundant hospital routers. The instructor pointed out that redundant routers alone do not protect against a physical connection failure. I helped update the design to propose primary and secondary WAN connections over physically diverse paths at each hospital, plus a proposed 5G backup connection if both terrestrial links fail.
- **Updated the network architecture diagram (Slide 4):** The diagram shows the three hospitals with dual routers and separate Clinical and Guest networks, the redundant ISP core and security perimeter (ASA1 and ASA0), the protected server network, the diverse WAN paths, the proposed 5G backup, and a conceptual Microsoft Azure EHR repository.
- **Provided failover evidence (Slides 7 and 10):** Supplied the NIH HSRP Active/Standby screenshots from my Week 2 OSPF/HSRP testing as evidence of simulated router redundancy.
- **Contributed to the Validation Status slide (Slide 10):** Helped organize our work into three categories (Demonstrated, Proposed Design, and Validate Before Cutover) so the client can see which claims are supported by testing and which still need production validation.
- **Reviewed the team master Packet Tracer file:** Reviewed the running configurations of all 17 routers, switches, and firewalls in the team master file (TeamEcho_Week2_Final18Oct.pkt) to confirm that the presentation matched the actual configuration, and documented gaps for the team (see Section 2).

- **Prepared the WAN and failover presenter content:** Under the team's presenter handoff plan, the Network Architect explains the WAN design and failover. I prepared talking points on equipment redundancy, diverse WAN paths, 5G backup, and the limits of our simulated tests.

Instructor Feedback and Design Changes

Feedback	Design change	Status
Redundant routers alone do not protect against physical connection failure	Primary and secondary WAN connections over physically diverse paths at each hospital	Proposed (conceptual in diagram)
Need connectivity if both terrestrial links fail	Proposed 5G backup connection at each hospital	Proposed (not tested)
Cloud storage protection needed more detail	Microsoft Azure EHR repository with AES-256 at rest, TLS 1.2+ in transit, Entra ID MFA, RBAC, audit logs, and tested backups	Proposed (teammate-led)

2. Diagrams and Visual Evidence

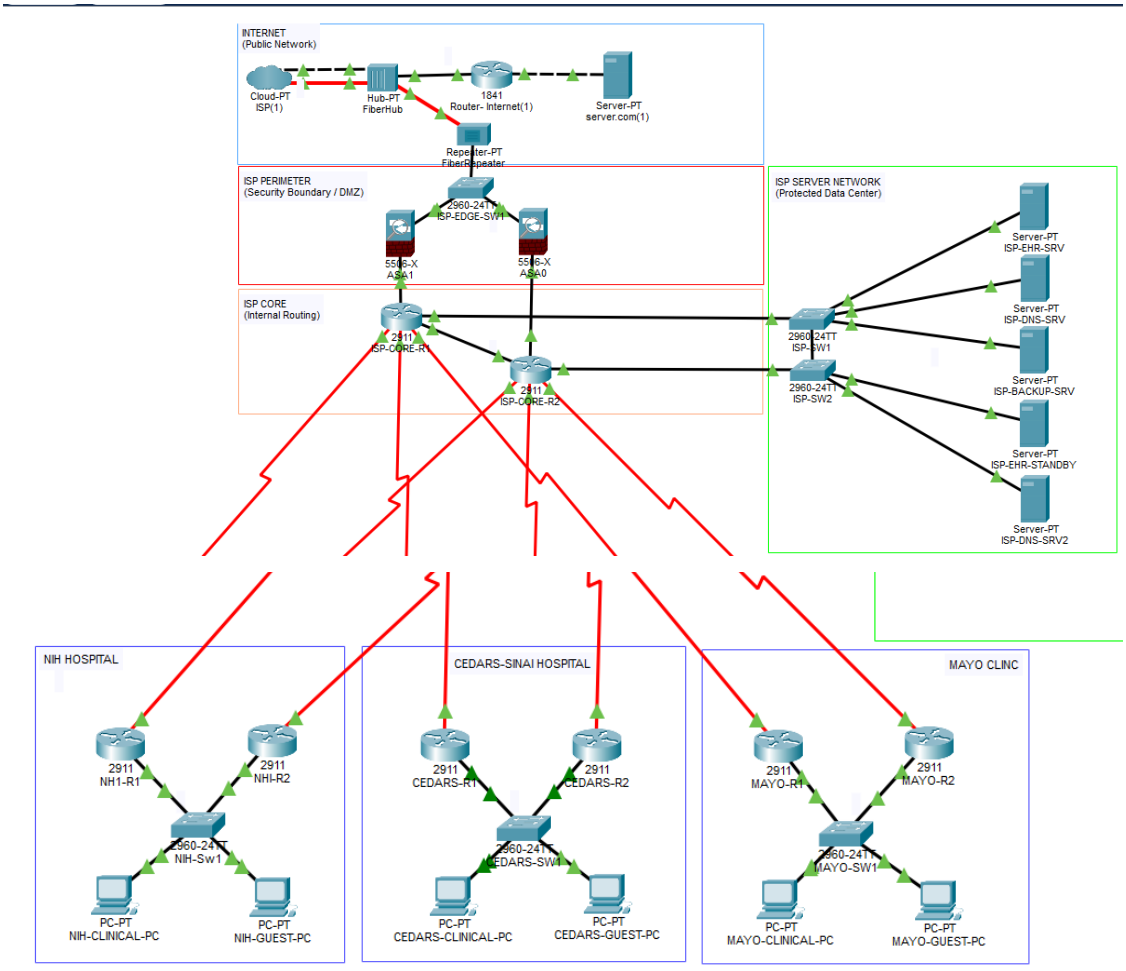


Figure 1. Team master Packet Tracer topology (saved October 18). Top: Internet, ISP perimeter with dual ASA firewalls (ASA1/ASA0), ISP core (ISP-CORE-R1/R2), and the protected server network (EHR, DNS, backup, standby EHR, secondary DNS). Bottom: NIH, Cedars-Sinai, and Mayo Clinic, each with dual routers, a shared switch, and Clinical and Guest PCs.

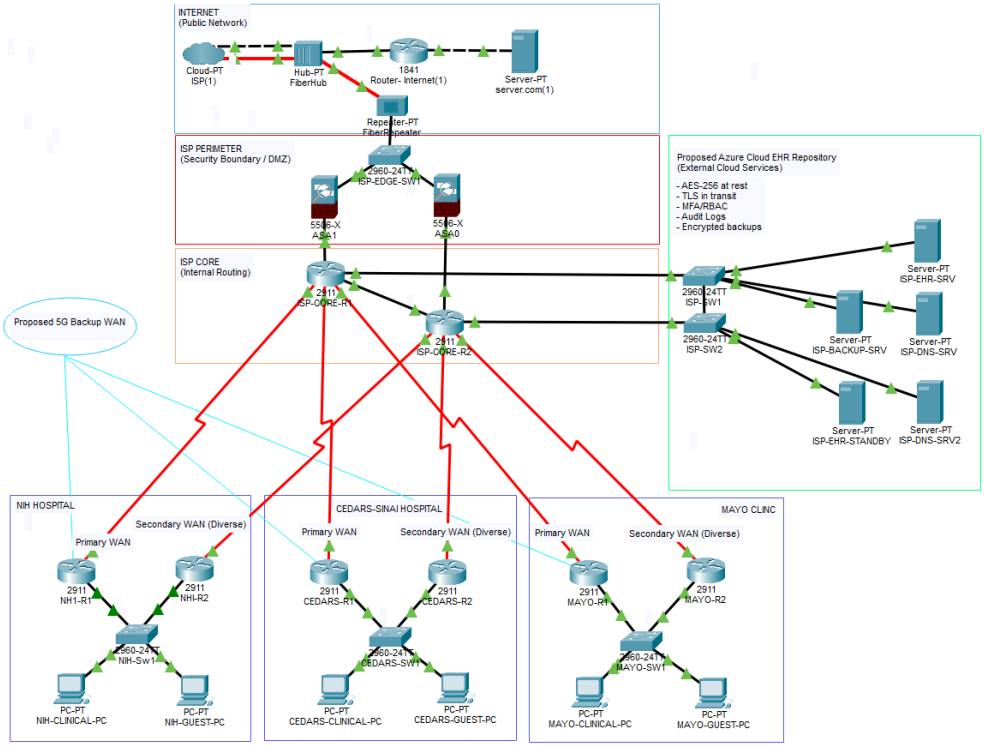


Figure 2. Updated architecture presented on Slide 4: physically diverse primary/secondary WAN paths, proposed 5G backup WAN, and a conceptual Azure cloud EHR repository.

```
NIH-R2>en
NIH-R2#sh standby brief
                     P indicates configured to preempt.
                     |
Interface   Grp  Pri P State    Active          Standby         Virtual IP
Gig         10   100 P Standby  10.20.10.2      local           10.20.10.1
Gig         30   100 P Standby  10.20.30.2      local           10.20.30.1
NIH-R2#
```

Copy Paste

Figure 3. NIH-R2 "show standby brief" output: Standby for HSRP groups 10 (Clinical) and 30 (Guest), with virtual gateways 10.20.10.1 and 10.20.30.1 and preemption enabled.

Validation Status Presented to the Client (Slide 10)

Demonstrated in simulation	Proposed design	Validate before cutover
Multi-site network mock-up	Physically diverse WAN paths	Circuit and device failover
Selected VLAN/ACL isolation tests	Independent 5G backup	EHR access and security
Simulated Internet access	Azure EHR repository	Backup restoration
NIH HSRP Active/Standby states	Cloud security and recovery	Monitoring and rollback

Configuration Review of the Team Master File

I compared the running configurations in the master file against the claims in our presentation. The review confirmed the main design, but also found gaps that the team should fix before the design is considered complete.

Area	What the configuration shows	Assessment
HSRP gateways	All three hospital R1 routers use priority 105 with preempt and track their primary serial WAN link; R2 routers use the default priority 100 and preempt.	Consistent at all three sites
Dynamic routing	OSPF runs only on NIH-R1, NIH-R2, both ISP core routers, and the Internet router. Cedars-Sinai and Mayo Clinic still rely on static routes plus floating static routes (AD 10) via the core-to-core link.	Partial – a LAN failure at Cedars-R1 or Mayo-R1 could leave a static route pointing at a failed path
Guest isolation	Each hospital's Guest ACL is applied inbound on VLAN 30 on both R1 and R2.	Isolation is preserved during router failover
Secondary DNS	Guest ACLs permit DNS only to 10.10.10.20 and deny 10.10.10.0/24; DHCP pools hand out only 10.10.10.20.	Gap – clients cannot fall back to secondary DNS 10.10.10.21
DHCP	DHCP pools exist only on the R1 router at each hospital.	Gap – if R1 fails, new clients cannot obtain addresses
Firewall path failover	ISP-CORE-R1's default route points only to ASA1 (10.10.254.2), with no floating default toward ISP-CORE-R2/ASA0.	Gap – traffic shifts to ASA0 only if the ASA1 link itself goes down
ASA0 vs. ASA1	Both firewalls have matching NAT and inside routes, but ASA0's inspection policy omits "inspect http," which ASA1 includes.	Lead to test for the open ASA0 return-traffic issue
Device hardening	No enable secret, console/VTY passwords, or SSH; unused switch ports are not shut down; no port security. NIH-R1, Cedars-R1, Mayo-R1, ISP-CORE-R1, and ASA1 keep default hostnames.	Gap – acceptable for a lab, not for production

3. Obstacles and How I Addressed Them

Obstacle	What happened	How I addressed it
Gap in WAN redundancy	Instructor feedback showed our router redundancy did not cover a physical circuit or connection failure.	Added physically diverse primary/secondary WAN paths and a proposed 5G backup to the architecture and the resiliency slide.
Packet Tracer limitations	Packet Tracer cannot realistically model carrier path diversity, 5G service, or Azure cloud services.	Showed these components conceptually in the diagram and stated in the speaker notes that they are proposals, not completed tests.
Risk of overstating results	Our ACL tests blocked selected Guest-to-Clinical traffic, and HSRP showed Active/Standby states, but neither proves full segmentation or circuit failover.	Used precise wording ("selected traffic," "demonstrated in simulation") and added the Validation Status slide.
Unresolved ASA0 return path	From Week 2, Internet return traffic through the secondary firewall ASA0 was still not validated.	Did not claim firewall failover in the presentation; listed device failover under "Validate Before Cutover." My configuration review found two leads: ASA0 lacks "inspect http," and ISP-CORE-R1 has no backup default route toward ASA0.
Detail vs. presentation time	The technical detail behind each design decision was too much for the slides.	Kept slides short and placed technical detail and limitations in the speaker notes and presenter guidance.

4. Resources and Their Evaluation

Resource	How it was used	Evaluation
NIST SP 800-66 Rev. 2, Implementing the HIPAA Security Rule	Safeguards for ePHI, risk assessment, and contingency planning	Authoritative federal guidance; highly relevant to a healthcare ISP
U.S. HHS, HIPAA Security Rule	Regulatory basis for administrative, physical, and technical safeguards	Primary regulatory source; authoritative
NIST SP 800-207, Zero Trust Architecture	Supported not trusting VPN or network location alone for EHR access	Authoritative; useful for access-control design
Microsoft Azure Architecture Center, Azure Storage security, Azure Backup docs	Encryption, access control, private access, and backup capabilities for the proposed repository	Vendor product guidance: accurate for features, but not evidence that anything was deployed or configured
Team master Packet Tracer file (saved Oct 18)	Running configurations of 17 routers, switches, and firewalls used to verify presentation claims	Most accurate source for what is actually configured; still a simulation
Team Packet Tracer topology and NIH validation evidence	Screenshots and results for the mock-up, isolation, and HSRP slides	Primary project evidence, but limited to a simulated environment
Team Echo Phase 1 Detailed Requirements Report	RFP requirements, sites, and expansion plans	Our own baseline; checked against the RFP

I relied on NIST and HHS for regulatory and security claims, used Microsoft documentation only to describe proposed Azure capabilities, and used our own Packet Tracer evidence only for what was actually demonstrated.

5. Legal, Ethical, and Security Considerations

Legal

- Using Azure does not make the environment HIPAA compliant by itself. The proposal requires a business associate agreement, HIPAA-eligible Azure services, a clear shared-responsibility split, and a formal risk assessment before handling ePHI.
- State requirements identified in Phase 1 (California’s Confidentiality of Medical Information Act and Minnesota’s Health Records Act) still apply to records from Cedars-Sinai and Mayo Clinic.
- The presentation does not claim HIPAA certification; the conceptual design alone is not a compliance determination.

Ethical

- Honest representation to the client: the presentation separates demonstrated results from proposed components so the client is not misled about readiness.
- No real patient data was used; all testing used simulated systems.
- Teammates’ work and outside sources are credited on the Sources slide.

Security

- **Availability:** Diverse WAN paths and 5G backup reduce single points of failure. Provider and building-entry diversity must be confirmed, because two circuits in the same conduit can fail together.
- **5G backup risk:** The 5G path must carry encrypted traffic and enforce the same firewall and access rules as the primary links; failover must not create a bypass around the security perimeter.
- **Segmentation:** Clinical VLAN 10 and Guest VLAN 30 with ACLs block selected Guest traffic to clinical systems. The final "permit ip any any" rule and Guest-to-Guest isolation still need review, and management/administrative VLANs are proposed for production.
- **EHR protection:** Proposed controls include Entra ID MFA, least-privilege RBAC, Azure Private Endpoints, AES-256 at rest, TLS 1.2+ in transit, audit logging, and tested restores.
- **Firewall redundancy:** ASA0 return traffic remains unvalidated, and ISP-CORE-R1 has no backup default route toward ASA0, so the secondary firewall path cannot yet be relied on during a primary firewall failure.
- **Service availability:** DHCP runs only on each hospital’s R1 router, and clients receive only the primary DNS server. A router or DNS failure could therefore interrupt access to EHR services even though the network paths remain up.
- **Device hardening:** The routers, switches, and firewalls have no enable secret, console or VTY passwords, or SSH, and unused switch ports are active. Production deployment needs SSH with AAA, strong credentials, disabled unused ports, port security, and a dedicated management VLAN.

6. Evaluation of Progress and Quality

RFP requirement	How the presentation addresses it	Current status
Secure Internet and EHR access	VLAN segmentation, firewalls, ACLs, encryption, Azure access controls	Partly demonstrated (selected ACL tests); Azure proposed
Redundant network connectivity	Dual routers, HSRP/OSPF, diverse WAN paths, 5G backup	NIH HSRP demonstrated; WAN diversity and 5G proposed
24/7 network operations	NOC monitoring, incident response, vulnerability management	Proposed
Scalable, compliant infrastructure	Cloud storage, expansion to Baltimore, Boston, Houston; HIPAA safeguards	Proposed; not a compliance certification

The presentation now covers all four key RFP requirements and responds directly to the instructor’s feedback on WAN redundancy and cloud storage. Its main strength is credibility: claims are tied to evidence, and limitations are stated openly rather than hidden.

Limits remain. Most resiliency improvements (diverse circuits, 5G, Azure) are conceptual. HSRP failover was demonstrated only at NIH, the ASA0 path is unresolved, and ICMP and ACL tests do not prove application availability or complete segmentation. The configuration review also showed that OSPF is configured only at NIH, DHCP and DNS lack redundancy, and the devices are not hardened. The design is a strong proposal, not a validated production network.

7. Next Steps

- Develop the detailed physical implementation plan: equipment, circuits, site requirements, and the phased NIH → Cedars-Sinai → Mayo Clinic rollout.
- Rehearse the WAN and failover segment and prepare answers on circuit diversity, 5G limitations, and HSRP/OSPF behavior.
- Extend OSPF to Cedars-Sinai and Mayo Clinic, then repeat the NIH failover tests at both sites.
- Test ASA0 with "inspect http" added and add a floating default route on ISP-CORE-R1 toward ASA0; retest firewall-path failover.
- Propose adding secondary DNS (10.10.10.21) to the DHCP pools and Guest ACLs, and DHCP on the R2 routers or a central DHCP server.
- Recommend basic hardening (hostnames, enable secret, SSH, shutting unused ports) before the final master file is submitted.
- Capture dated screenshots of the updated topology and add them to the cumulative journal.

8. References

- National Institute of Standards and Technology. (2020). *Zero trust architecture (NIST SP 800-207).* https://doi.org/10.6028/NIST.SP.800-207
- National Institute of Standards and Technology. (2024). *Implementing the Health Insurance Portability and Accountability Act (HIPAA) Security Rule: A cybersecurity resource guide (NIST SP 800-66 Rev. 2).* https://doi.org/10.6028/NIST.SP.800-66r2
- Team Echo. (2026). *Healthcare ISP detailed requirements report and Cisco Packet Tracer project materials* [Unpublished course project].
- Team Echo. (2026). *Healthcare ISP RFP presentation* [Unpublished course presentation].

CYB495 – Information and Cybersecurity Capstone

4.2 Assignment: Week 4 Journal – Physical Implementation

Student	Ernest Foubi
Team	Team Echo (Ernest Foubi and three teammates)
Role	Network Architect/Engineer
Project	Healthcare ISP Network Infrastructure and Security Architecture
Week	Week 4 – Physical Implementation

1. Contributions to the Project

During Week 4, I focused on reviewing and improving Team Echo's Physical Implementation Plan for our proposed Healthcare ISP network connecting the National Institutes of Health (NIH), Cedars-Sinai, and Mayo Clinic. As the Network Architect, I reviewed and updated the equipment requirements, WAN connectivity design, implementation assumptions, and cost-planning sections.

- **Greenfield deployment assumption:** One of my main contributions was clarifying that our cost estimate assumes a complete new network infrastructure deployment, also known as a greenfield deployment, rather than reusing existing networking equipment.
- **Equipment quantities:** I helped establish the proposed network equipment quantities: two WAN routers, two firewalls, and two managed switches at each hospital and at the ISP core (eight of each), plus one 5G device per hospital.
- **WAN design:** I reviewed the design for primary and secondary WAN connections with 5G backup connectivity. Our proposal includes six terrestrial WAN connections and three 5G backup services across the three hospitals.
- **Budget review:** I reviewed the implementation budget and identified an inconsistency in the annual recurring costs. Based on the proposed monthly expense of $31,600, the correct annual operating estimate is $379,200. I communicated this finding to the team for correction, and the final plan now shows $379,200.

Key Figures in the Final Plan

Item	Final plan figure
Terrestrial WAN circuits (primary + secondary)	6 (2 per hospital)
5G backup services	3 (1 per hospital)
Total equipment cost	$164,000
One-time implementation cost	$280,000
Monthly recurring cost	$31,600
Annual recurring cost (corrected)	$379,200 ($31,600 × 12)
First-year planning total	$659,200

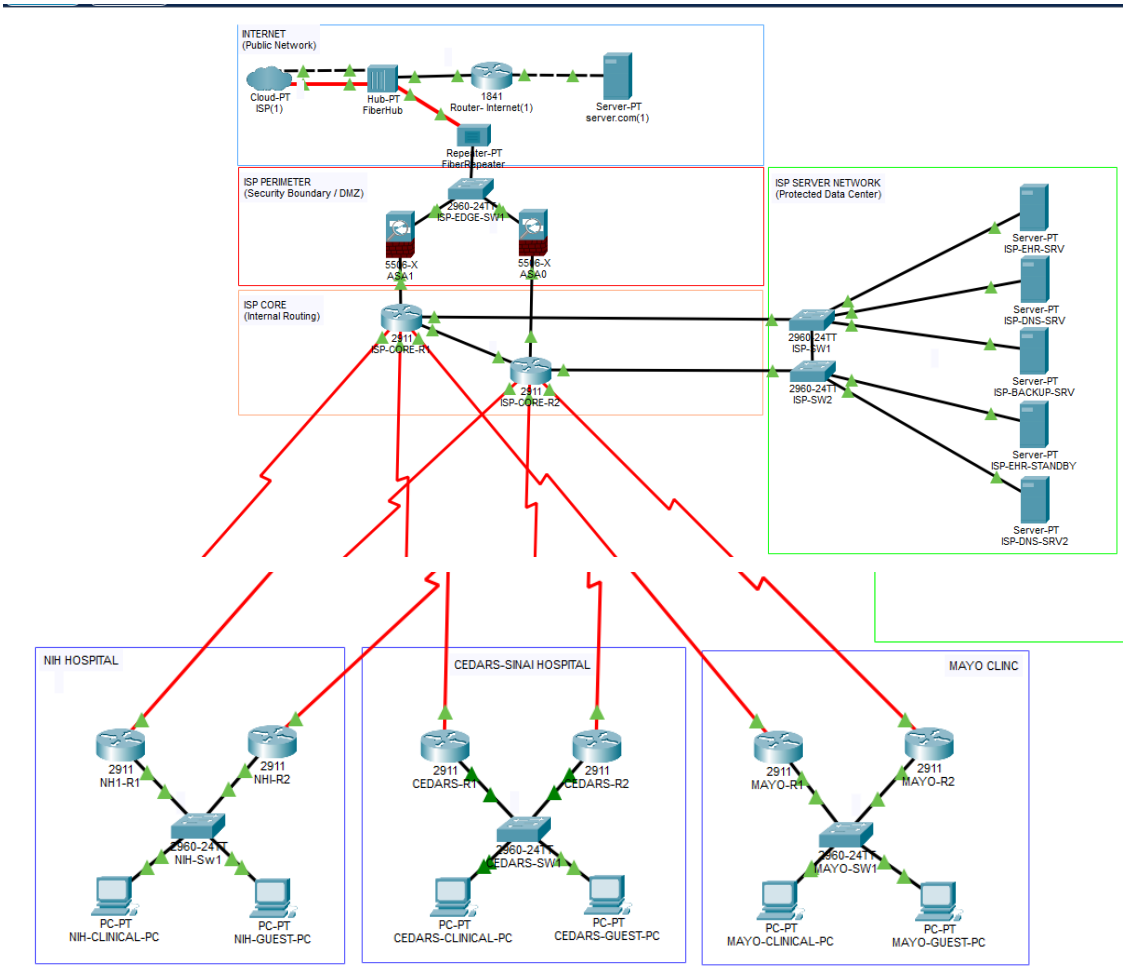
2. Obstacles Encountered and Solutions

Obstacle	What happened	How I addressed it
Inconsistent equipment-reuse assumption	Some sections initially suggested reusing existing equipment, while our proposed budget was intended to represent a completely new deployment.	I revised the relevant network architecture sections to establish a consistent greenfield deployment assumption.
WAN redundancy	Simply installing multiple routers does not guarantee network availability if the connections depend on the same physical infrastructure.	I recommended separate primary and secondary WAN connections, preferably using different service providers and physically diverse paths, with 5G as an additional backup option.
Accuracy of cost estimates	The annual recurring figure did not match the monthly total, and equipment prices were preliminary.	I recommended verifying equipment quantities, reviewing UPS redundancy requirements, and checking vendor pricing before finalizing the budget.

3. Visual Documentation and Supporting Evidence

I used our Cisco Packet Tracer topology and the Physical Implementation Plan to support the proposed infrastructure design. The network diagram illustrates the centralized ISP core and connections to NIH, Cedars-Sinai, and Mayo Clinic. Each hospital includes two routers, a switch, and separate Clinical and Guest workstations. Our earlier Packet Tracer testing demonstrated basic WAN connectivity, static routing, VLAN configuration, and reachability to the centralized EHR server.

Figure 1 – Healthcare ISP Network Topology



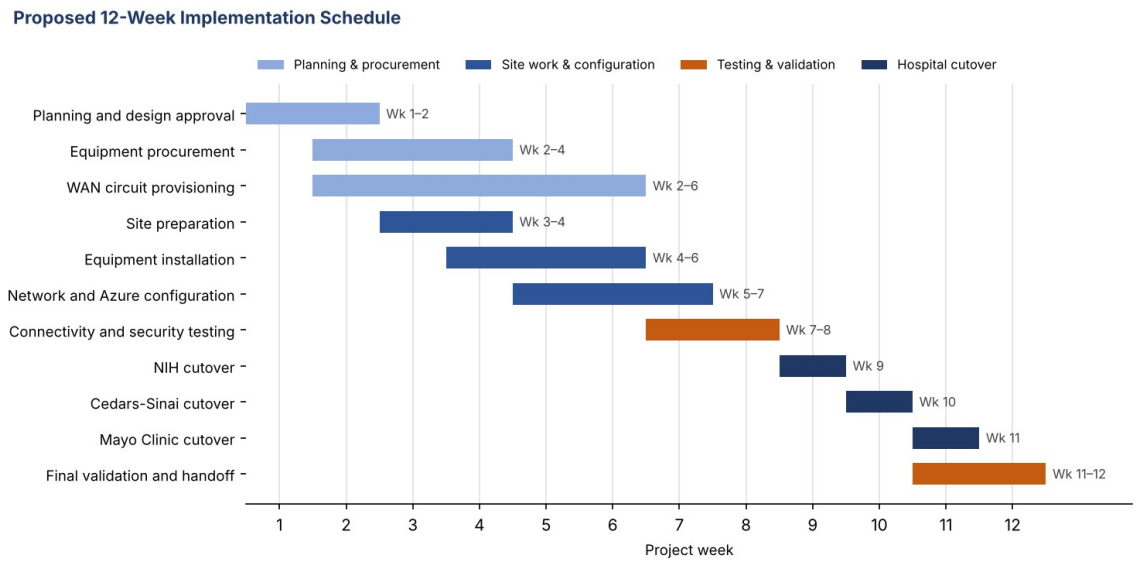
Cisco Packet Tracer topology: Internet, ISP perimeter with dual ASA firewalls, ISP core, protected server network, and NIH, Cedars-Sinai, and Mayo Clinic with Clinical and Guest workstations.

Figure 2 – Proposed Equipment Requirements and Cost Estimates

Equipment	Quantity	Unit Cost	Estimated Total
Enterprise WAN routers	8	$5,500	$44,000
Next-generation firewalls	8	$6,500	$52,000
Managed network switches	8	$2,800	$22,400
5G-capable WAN routers/modems	3	$1,200	$3,600
Rack equipment and accessories	4	$2,500	$10,000
UPS / backup power equipment	4	$3,500	$14,000
Cabling and transceivers	1	$18,000	$18,000
Total Equipment Cost			$164,000

Source: Table 1, Team Echo Physical Implementation Plan. Prices are illustrative planning allowances, not vendor quotations.

Figure 3 – WAN Redundancy and Implementation Planning



Proposed 12-week schedule from the Physical Implementation Plan (Table 3). WAN circuit provisioning runs in Weeks 2–6, and cutover is staggered: NIH (Week 9), Cedars-Sinai (Week 10), and Mayo Clinic (Week 11).

WAN implementation requirements from the plan:

- Install primary and secondary WAN circuits at each hospital.
- Verify physical route diversity and provider independence.
- Configure routing and automatic failover between connections.
- Install and configure the 5G backup equipment.
- Establish secure VPN connectivity to Azure over the backup connection.
- Test each connection independently before production cutover.

These materials document the progression from our simulated network architecture to the proposed production implementation.

4. Resources Used

The primary resources used during Week 4 included Team Echo’s Physical Implementation Plan, the project RFP requirements, instructor feedback, our Cisco Packet Tracer topology, and the team’s implementation worksheet. I also used the following authoritative technical references to support the design review:

Resource	Usefulness	Accuracy / limitations
Cisco Catalyst 8200 Series documentation	Useful for evaluating enterprise WAN routing capabilities and equipment specifications.	Actual equipment selection and pricing require vendor verification.
Microsoft Azure Pricing Calculator	Useful for estimating cloud infrastructure expenses.	Final costs depend on selected services, usage, and contractual pricing.
NIST SP 800-34 Rev. 1	Provides guidance on contingency planning, recovery strategies, and information system availability.	Authoritative federal guidance; must be adapted to the hospitals' requirements.
HHS HIPAA Security Rule guidance	Helps identify administrative, physical, and technical safeguards relevant to ePHI.	Authoritative regulatory source; does not replace a formal compliance assessment.

These sources support our proposed architecture but do not replace site surveys, vendor quotations, or formal compliance assessments.

5. Legal and Ethical Considerations

Because our proposed network supports healthcare organizations, protecting electronic protected health information (ePHI) is an essential responsibility.

- **Legal:** The HIPAA Security Rule establishes requirements for safeguarding ePHI, including appropriate administrative, physical, and technical protections. Our proposal addresses these concerns through controlled access, network segmentation, encryption, monitoring, and disaster recovery planning. The plan also requires a business associate agreement with the cloud provider before patient data is stored in Azure.
- **Ethical:** From an ethical perspective, the network must support patient confidentiality while ensuring that authorized healthcare professionals can access essential information when needed. The plan keeps the existing hospital Internet connection available until the new network passes acceptance testing, so cutover does not disrupt patient care.

- **Accurate representation:** Another important responsibility is accurately representing the project's capabilities. Our Packet Tracer simulation demonstrates selected networking functions, but it does not establish that the proposed production environment is fully secure or HIPAA-compliant. Cost figures are likewise labeled as planning estimates, not confirmed prices.

6. Security Considerations

Security was an important factor throughout my review of the Physical Implementation Plan.

- **Segmentation:** Our network design uses VLAN 10 for Clinical users and VLAN 30 for Guest users at each hospital. However, VLAN separation alone is insufficient to prevent unauthorized access. The production implementation requires appropriate firewall policies, access control lists, and network monitoring.
- **WAN security and availability:** For WAN security, I recommended encrypted connections between healthcare locations and centralized EHR services. The proposed redundant WAN architecture also supports availability by reducing dependence on a single connection.
- **Data protection:** Additional proposed safeguards include multifactor authentication, role-based access control, encryption at rest and in transit, audit logging, protected backups, and disaster recovery procedures.
- **Validation before deployment:** Before deployment, the team would need to validate guest isolation, authorized EHR access, WAN failover, backup restoration, and security monitoring.

Validation test	Pass criterion (from the plan)	Owner
VLAN segmentation	Guest VLAN cannot reach clinical networks	Security team
Primary WAN failover	Secondary link carries authorized traffic within approved threshold	Network team
5G backup	Critical EHR traffic reaches Azure over encrypted VPN	WAN team
Firewall and access	Unauthorized traffic denied; MFA and TLS verified	Security team
Recovery	Test data restored within approved RTO/RPO	Cloud team

7. Evaluation of Progress and Project Quality

I believe Team Echo made significant progress during Week 4 by developing a more complete Physical Implementation Plan that includes network equipment, installation activities, WAN connectivity, cloud services, security controls, disaster recovery, and implementation costs.

My contributions improved the consistency between the network architecture and the financial assumptions. Clarifying the greenfield deployment approach also helped establish a more realistic basis for estimating equipment and installation expenses.

Our collaboration remained important throughout this phase. The project lead coordinated the implementation plan and final document revisions, the security lead reviewed security requirements and supporting documentation, the cloud lead expanded the Azure infrastructure and disaster recovery sections, and I focused on network architecture and cost validation.

Area	Assessment
Completed	Equipment list, WAN design, installation process, 12-week schedule, testing and cutover plan, cost estimates
Improved this week	Consistent greenfield assumption; corrected annual recurring cost
Still preliminary	Equipment prices, WAN provisioning timelines, and some redundancy requirements (e.g., UPS runtime)

Although the proposal is substantially developed, the equipment prices remain preliminary, WAN provisioning timelines require provider confirmation, and some redundancy requirements need further engineering validation.

Overall Assessment: The Week 4 deliverable provides a more complete foundation for the final Healthcare ISP proposal. My next responsibility is to support the final presentation by clearly explaining our network architecture, equipment requirements, connectivity strategy, and proposed implementation approach.

References

- Cisco Systems. (n.d.). Cisco Catalyst 8200 Series Edge Platforms data sheet. https://www.cisco.com/c/en/us/products/collateral/routers/catalyst-8200-series-edge-platforms/nb-06-cat8200-series-edge-plat-ds-cte-en.html
- Microsoft Azure. (n.d.). Pricing calculator. https://azure.microsoft.com/en-us/pricing/calculator/
- National Institute of Standards and Technology. (2010). Contingency planning guide for federal information systems (NIST SP 800-34 Rev. 1). https://doi.org/10.6028/NIST.SP.800-34r1
- U.S. Department of Health and Human Services. (n.d.). Summary of the HIPAA Security Rule. https://www.hhs.gov/hipaa/for-professionals/security/laws-regulations/index.html
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Week 5 Journal – Presenting the Proposal

Student	Ernest Foubi
Team	Team Echo (Ernest Foubi and three teammates)
Role	Network Architect/Engineer
Week	Week 5 – Presenting the Proposal

1. Contributions to the Project

During Week 5, Team Echo presented its final 21-slide healthcare ISP proposal, with each team member presenting a specific portion. The project lead presented the introduction, RFP requirements, and architecture overview (slides 1–5); the security lead presented security, compliance, and EHR protection (slides 6–10); the cloud lead presented infrastructure, deployment, and testing (slides 11–15); and I presented slides 16–19.

- **Presented my portion of the proposal:** I presented and recorded Cloud EHR Disaster Recovery (slide 16), the One-Time Implementation Budget (slide 17), Operating Costs and First-Year Outlook (slide 18), and the Closing Recommendation (slide 19). I recorded my portion after the cloud lead so the presentation flowed from deployment and testing into recovery and costs.
- **Connected my slides to the network design:** Although these slides focus on recovery and finance, I tied them to the network architecture. For example, I explained that WAN redundancy and cloud recovery solve different failures: diverse WAN paths keep a hospital connected, but they do not restore a failed EHR application or corrupted records.
- **Final figure check:** Before recording, I verified that every total on my slides reconciles with its line items (Table 1) and that the corrected annual recurring cost of $379,200 from Week 4 is now used consistently.
- **Prepared to explain budget changes:** The final deck's one-time estimate ($262,000) is lower than the Phase 4 plan ($280,000) because the team removed the duplicated $18,000 cabling allowance and moved separate site-preparation costs to "to be determined." I prepared my narration to explain this difference.

Table 1. Cost Figures Verified for Slides 17 and 18

Figure	Calculation	Result
One-time implementation	$164,000 + $30,000 + $24,000 + $32,000 + $12,000	$262,000
15% upfront contingency	$262,000 × 0.15	$39,300
WAN and 5G (monthly)	$7,500 + $4,500 + $600	$12,600
Azure cloud services (monthly)	$9,000 + $2,500 + $1,800	$13,300
Monitoring, support, licensing (monthly)	$3,200 + $2,500	$5,700
Total monthly recurring	$12,600 + $13,300 + $5,700	$31,600
Annual recurring	$31,600 × 12	$379,200
Year one (before contingency/taxes)	$262,000 + $379,200	$641,200

2. Obstacles and Approaches

Obstacle	Why it mattered	My approach
Slides outside my primary role	My assigned slides covered disaster recovery and finance rather than network architecture.	I studied recovery time and recovery point objectives (RTO/RPO) and linked each slide back to network availability, so I could present the material accurately.
Budget differences between documents	The Phase 4 plan ($280,000) and the final deck ($262,000) showed different one-time totals, which could confuse the audience.	I reconciled the figures and prepared a clear explanation: the duplicated cabling allowance was removed and site preparation is still TBD.
Risk of overstating recovery capability	A 15-minute backup interval alone does not guarantee a 15-minute RPO, and Azure recovery has not been tested.	I presented the RTO, RPO, and retention values as proposed targets that require approval and acceptance testing.
Coordinating four presenters	Separate recordings can produce gaps or repeated content between sections.	I confirmed slide assignments with the team and recorded after the cloud lead so my section began where the testing content ended.

3. Visual Documentation

The figures below are the four slides I presented in the final proposal.

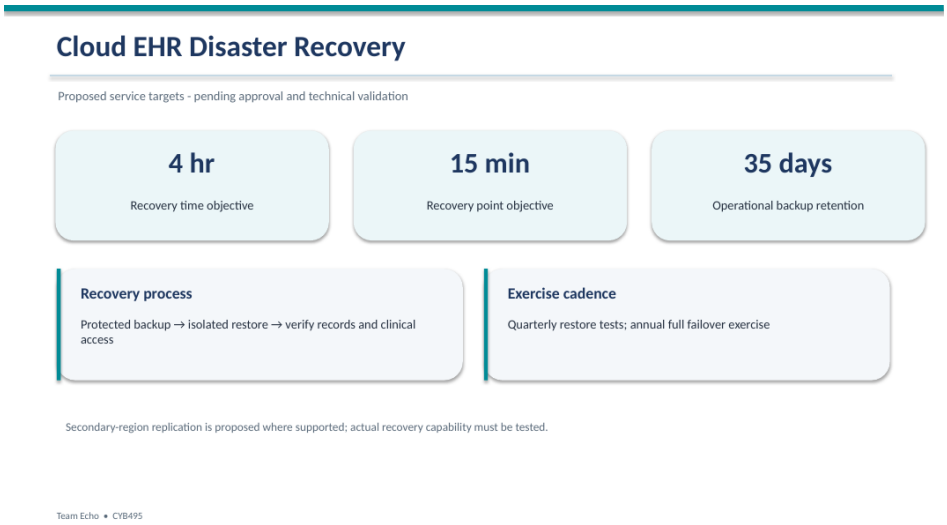


Figure 1. Slide 16 – Cloud EHR Disaster Recovery: proposed 4-hour RTO, 15-minute RPO, and 35-day backup retention.

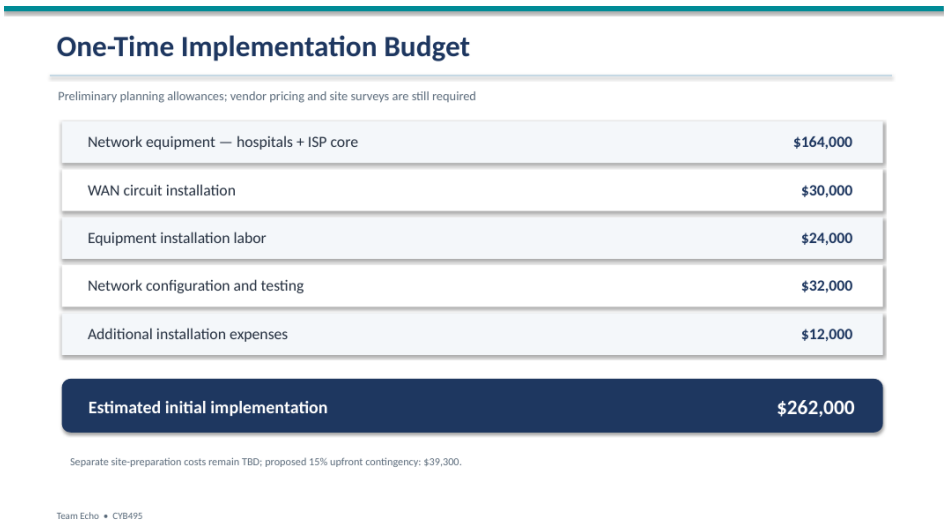


Figure 2. Slide 17 – One-Time Implementation Budget: $262,000 estimate with a proposed 15% contingency of $39,300.



Figure 3. Slide 18 – Operating Costs and First-Year Outlook: $31,600 per month, $379,200 per year, and $641,200 for year one.

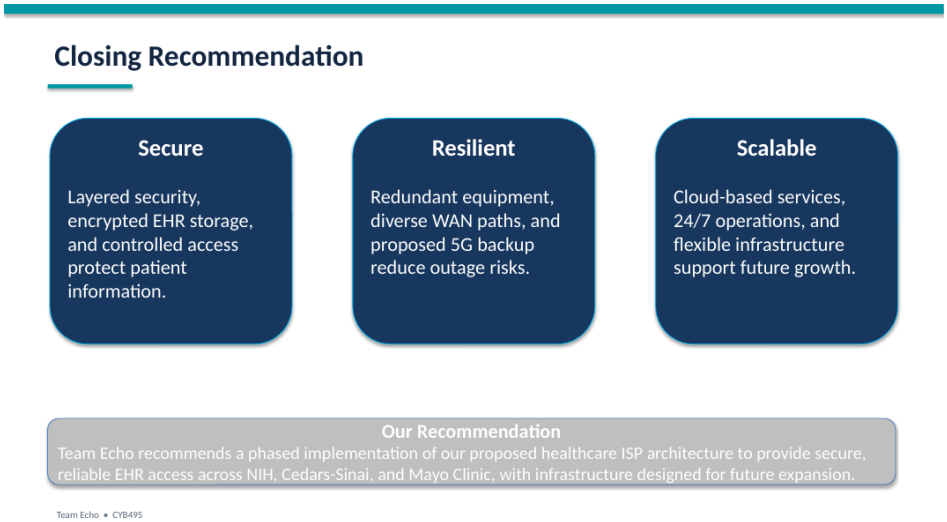


Figure 4. Slide 19 – Closing Recommendation: a phased, secure, resilient, and scalable implementation.

4. Resources Used

Resource	How I used it	Evaluation
NIST SP 800-34 Rev. 1, Contingency Planning Guide	Definitions and planning approach for RTO, RPO, backup, and recovery testing	Authoritative federal guidance; highly relevant to slide 16
HHS, Summary of the HIPAA Security Rule	Contingency plan and data backup requirements for ePHI	Primary regulatory source; authoritative
NIST SP 800-66 Rev. 2, Implementing the HIPAA Security Rule	How HIPAA safeguards apply to a healthcare network	Authoritative and current (2024)
Microsoft Azure Pricing Calculator	Basis for the Azure cost allowances on slide 18	Official tool, but estimates depend on sizing inputs we have not finalized
Team Echo Phase 4 Physical Implementation Plan and Phase 5 deck	Source of all cost and recovery figures	Primary project sources; I verified their arithmetic
Instructor feedback (the course instructor)	His question about a design flaw found at NIH shaped the cutover and rollback approach	Directly relevant; improved the realism of the plan

5. Legal and Ethical Considerations

- **HIPAA contingency planning:** The HIPAA Security Rule requires covered entities to have a contingency plan, including a data backup plan, a disaster recovery plan, and an emergency mode operation plan. Slide 16 addresses these requirements through protected backups, an isolated restore process, and scheduled recovery exercises.
- **Business associate responsibilities:** Storing ePHI in Azure requires a business associate agreement and HIPAA-eligible services. Using a cloud provider does not by itself make the environment compliant.
- **Honest cost representation:** All budget figures were presented as preliminary planning allowances, not vendor quotations. Presenting them as firm prices could mislead the client into an unrealistic budget.
- **Patient safety over schedule:** The team's plan holds the next hospital cutover if testing fails, rather than risking patient care to meet the 12-week target.
- **Credit and accountability:** Each team member presented and was credited for their own portion, and the deck cites all external sources.

6. Security Considerations

- **Protected backups:** Backups must be encrypted and access-controlled so that ransomware or a compromised account cannot delete or alter them.
- **Isolated restore and integrity checks:** Restores are performed in an isolated environment, and patient records are verified for integrity before clinical access is restored.
- **Access during recovery:** Multifactor authentication and role-based access control must still apply after failover, so recovery does not open a bypass to patient data.
- **WAN redundancy vs. cloud recovery:** Diverse WAN paths and 5G backup protect connectivity; cloud recovery protects the EHR application and data. Both are needed for availability.
- **Rollback path:** The legacy ISP connection remains available until each hospital signs off, so a failed cutover can be reversed without losing access.

7. Evaluation of Progress and Quality

Team Echo completed a full proposal that addresses all four key RFP requirements: secure Internet and EHR access, redundant connectivity, 24/7 operations, and scalable, compliant infrastructure. My slides presented recovery and cost information that was internally consistent and clearly labeled as proposed.

Area	Assessment
Strengths	Consistent and verified cost figures; clear distinction between tested and proposed capabilities; realistic, phased cutover with rollback
Limitations	Costs are planning estimates; Azure recovery, WAN diversity, and 5G failover have not been tested; the $3,200 monthly monitoring allowance may not cover a fully staffed 24/7 NOC
My performance	I delivered my assigned slides accurately, although they were outside my main network role; presenting networking slides would have used my Packet Tracer work more directly

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Week 5 Addendum – Packet Tracer Corrections and Validation

Student	Ernest Foubi
Date	November 5, 2026
File	Foubi_HealthcareISP_Corrected.pkt (my own copy of the team master file TeamEcho_Week2_Final18Oct)
Purpose	Apply and test the corrections identified in my Week 3 configuration review

1. Contributions

In Week 3, my review of the team master Packet Tracer file found that OSPF ran only at NIH, the secondary firewall (ASA0) path was unvalidated, Guest ACLs blocked the secondary DNS server, and DHCP ran only on the R1 routers. After the final presentation, I applied and tested the fixes in my own copy of the file, so the team master file remained unchanged.

Table 1. Corrections Applied and Test Results

Correction	What I configured	Test result
1. OSPF at all hospitals	OSPF on CEDARS-R1/R2 and MAYO-R1/R2 (passive LAN subinterfaces); added WAN networks on both cores; removed 15 static routes that overrode OSPF	ISP-CORE-R1 and ISP-CORE-R2 each have 4 FULL neighbors (previously 2); hospital networks learned as OSPF routes
1a. Cedars-Sinai failover test	Shut down CEDARS-R1 Gi0/0	CEDARS-R2 became HSRP Active; core rerouted 10.30.10.0/24 via 172.16.0.14 (metric 66); Clinical PC reached the EHR server (3/4 replies)
2. ASA0 backup path	Floating default routes (AD 20) on both cores; added "inspect http" to ASA0	With ASA1's link shut down: backup route [20/0] became active; ASA0 translated NIH traffic (show xlate); ping to server.com 3/4; server.com web page loaded
3. Guest ACLs	Rebuilt all six Guest ACLs (13 entries): allow DNS to 10.10.10.20 and 10.10.10.21; deny other Guest networks, WAN links, and firewall links	Guest PCs at all three hospitals blocked from the EHR server, other Guest networks, and WAN links; Internet access still works; ACL counters show matches on the new deny entries
4. Backup DHCP	R1 serves .21–.150; R2 serves .151–.254 with the HSRP gateway (.1)	With NIH-R1's LAN down, NIH-CLINICAL-PC received 10.20.10.151 from NIH-R2 and reached the EHR server (4/4 replies)

2. Obstacles and Approaches

Obstacle	What happened	How I addressed it
Commands pasted into the wrong router	Twice, the CEDARS-R1 OSPF commands were pasted into the ISP-CORE-R1 window that was still open. This added a duplicate router ID and two incorrect networks.	I removed the incorrect lines and verified the OSPF section. I then adopted an identity check: before every paste, run "show ip interface brief" and confirm the device's unique IP address.

Obstacle	What happened	How I addressed it
Accidental shutdown of the ASA1 link	During the Cedars-Sinai test, a "shutdown" also went into ISP-CORE-R1, disabling Gi0/0 (the link to ASA1).	Verification output showed the interface as administratively down; I restored it with "no shutdown" and confirmed it was up/up before continuing.
Static routes overriding OSPF	Static routes (AD 1 and 10) always beat OSPF (AD 110), so enabling OSPF alone changed nothing.	Enabled OSPF at the hospitals first, then removed the conflicting static routes on the cores, keeping the Internet default route and the server-network backup.
Proving failover, not just configuration	HSRP takeover already worked before the fix, so it did not prove the OSPF correction.	I captured the core routing table (metric 66 via the other core) and an end-to-end ping during each failure.
First-packet timeouts	The first ping after a failover often timed out.	I repeated the test; one lost packet during path convergence and ARP resolution is expected.

3. Visual Evidence

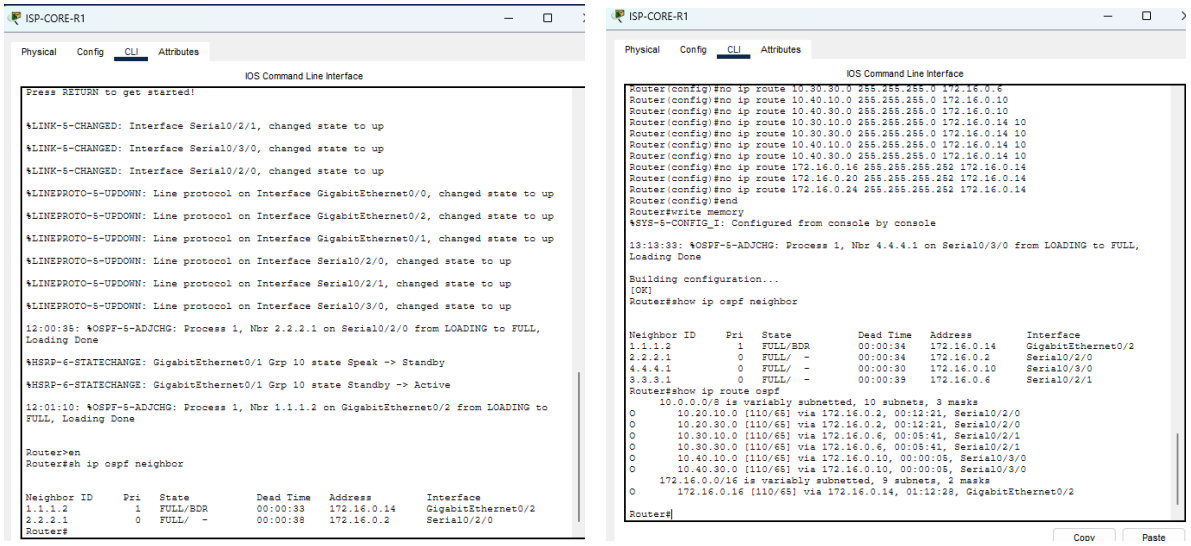


Figure 1. ISP-CORE-R1 before (left: 2 OSPF neighbors) and after (right: 4 neighbors, with Cedars-Sinai and Mayo Clinic networks learned through OSPF).

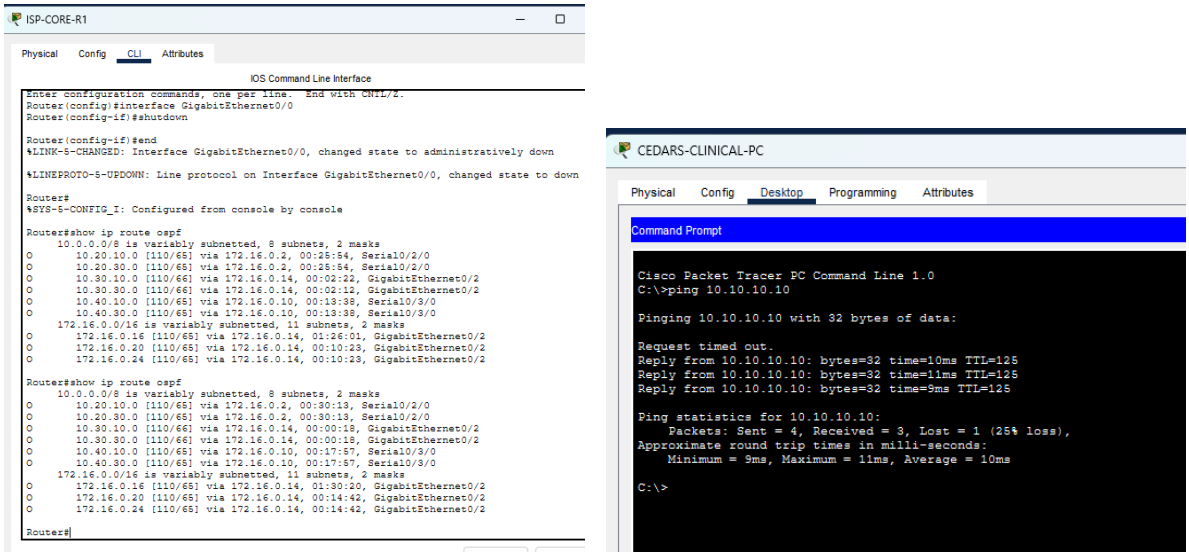


Figure 2. Cedars-Sinai failover: ISP-CORE-R1 reroutes 10.30.10.0/24 via ISP-CORE-R2 (left), and CEDARS-CLINICAL-PC still reaches the EHR server (right).

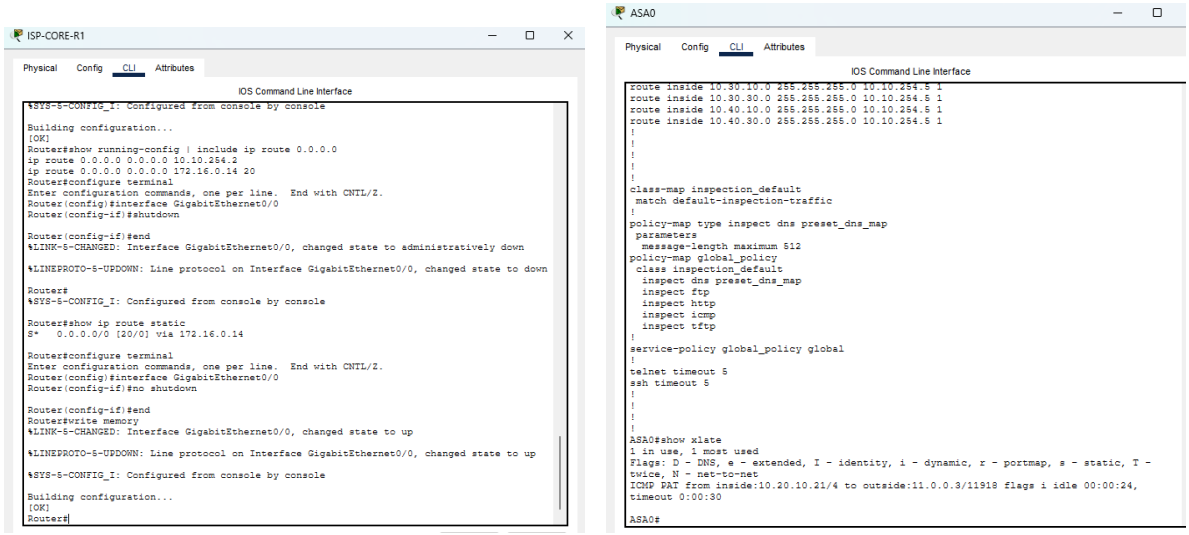


Figure 3. ASA0 failover: with the ASA1 link down, the floating default route [20/0] becomes active (left), and ASA0 translates the NIH PC's traffic (right).

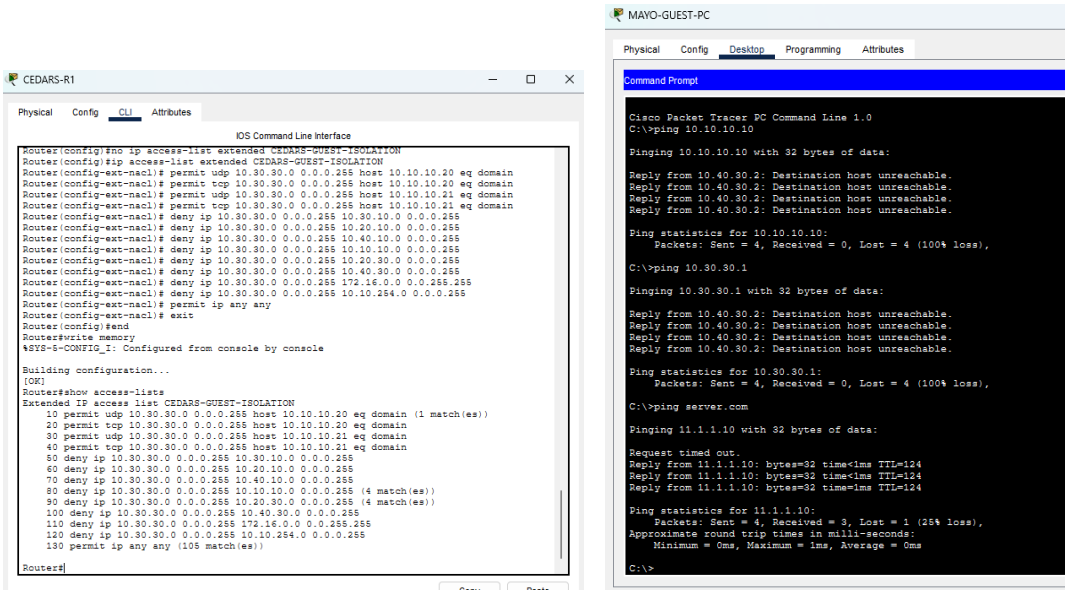


Figure 4. Rebuilt Guest ACL on CEDARS-R1 with match counters on the new deny entries (left), and MAYO-GUEST-PC blocked from the EHR server and another hospital's Guest network while Internet access works (right).

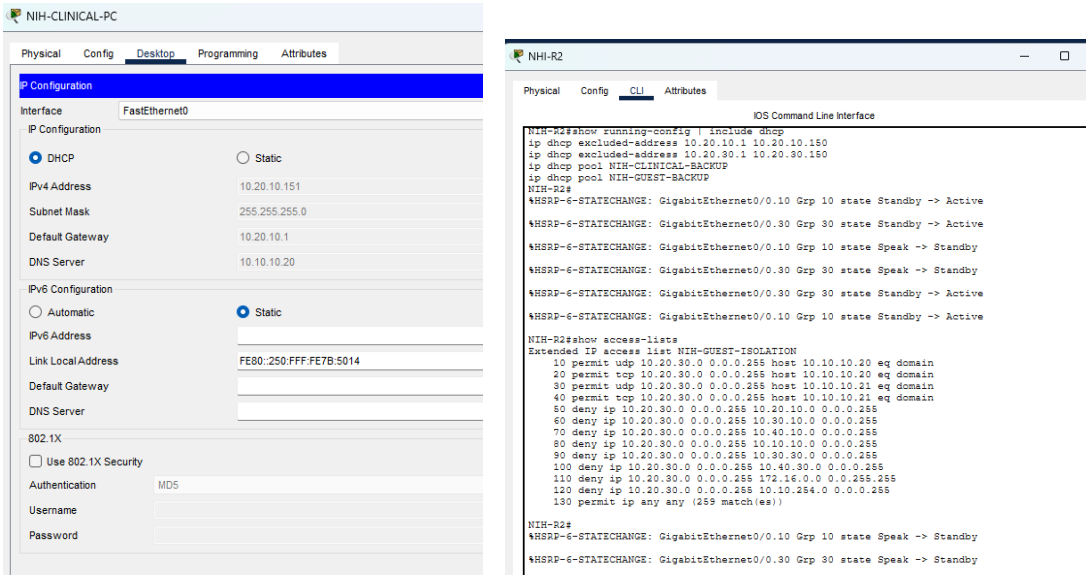


Figure 5. Backup DHCP: NIH-CLINICAL-PC receives 10.20.10.151 from NIH-R2 while NIH-R1 is down (left); NIH-R2 backup pools and 13-entry Guest ACL (right).

4. Resources

Resource	How I used it	Evaluation
My Week 3 configuration review and correction guide	List of gaps and the exact commands for each device	Built from the actual running configurations; accurate for this file
Cisco IOS command output (show ip ospf neighbor, show ip route, show standby brief, show access-lists, show xlate)	Evidence that each change worked	Most reliable source: shows the device's real state rather than intended configuration
Cisco Packet Tracer simulation	Test environment for failure scenarios	Useful for routing and ACL behavior; does not model real circuits, timing, or production load

5. Legal and Ethical Considerations

- **Respect for teammates' work:** I made all changes in my own copy of the file, not in the project lead's master file, and will share the results with the team before any merge.
- **Accurate reporting:** I reported my configuration mistakes and how I corrected them, and I only describe as "tested" the scenarios I actually ran.
- **Healthcare context:** Guest isolation and backup DHCP support HIPAA's technical safeguards for access control and availability, but a simulation does not establish compliance.

6. Security Considerations

- **Tighter Guest isolation:** Guests can no longer reach other hospitals' Guest networks, WAN links, or firewall links, which reduces lateral movement and reconnaissance from untrusted devices.
- **Availability:** OSPF failover at every hospital, a backup firewall path, a reachable secondary DNS server, and backup DHCP remove single points of failure identified in Week 3.
- **Remaining gap – device hardening:** Devices still have no enable secret, SSH, or port security, and some keep default hostnames. Hardening was deferred so the team and instructor can still open and verify the devices; for production, SSH with AAA, strong credentials, port security, and disabled unused ports are required.

7. Evaluation

The corrections resolved four of the five gaps from my Week 3 review, including the ASA0 issue that had remained open since Week 2. Every fix was verified with command output and an end-to-end test rather than configuration alone. The main limitations are that failover was demonstrated at one site per scenario (Cedars-Sinai for OSPF and NIH for DHCP and ASA0), device hardening remains outstanding, and Packet Tracer cannot validate real circuit diversity, 5G, or Azure services.

Final Reflection and Overall Evaluation

How the Solution Developed

The project progressed from requirements analysis (Week 1) to a working Packet Tracer mock-up (Week 2), a revised architecture presented to the client (Week 3), a physical implementation plan with schedules and costs (Week 4), and a final proposal presentation (Week 5). The design matured significantly along the way. Our first design relied mainly on redundant routers; after instructor feedback, it included physically diverse WAN circuits, 5G backup, a dual-firewall perimeter, and cloud disaster recovery.

Overall Quality of the Proposed Solution

Area	Assessment
Demonstrated in simulation	Multi-site connectivity; Clinical/Guest segmentation with tightened ACLs; OSPF and HSRP failover (NIH and Cedars-Sinai); backup firewall path through ASA0; backup DHCP
Proposed but not yet validated	Diverse WAN circuits, 5G emergency backup, and Azure EHR security and recovery
Known gaps	Four of five Week 3 gaps corrected and tested in my copy of the file (Week 5 Addendum); device hardening (SSH, passwords, port security) remains
Overall	A strong, realistic proposal that clearly separates evidence from plans; not a production-ready or HIPAA-certified network

Lessons Learned

- **Evidence matters more than appearance:** A topology that looks complete is not proof that it works. Comparing configurations and test results against our claims revealed gaps we would otherwise have presented as finished.
- **Redundancy must be physical, not only logical:** Two routers sharing one circuit or one conduit still have a single point of failure.
- **Verify before you change:** Pasting commands into the wrong router taught me to confirm a device's identity with a unique IP address before every change, and to check results immediately.
- **Check the numbers:** Small arithmetic errors, such as the annual recurring cost, can undermine an otherwise credible proposal.
- **Respect teammates' work:** When I found problems in shared files, I reported them and coordinated changes instead of editing a teammate's master configuration without approval.
- **Documentation saves time:** Keeping dated evidence each week made it much easier to build the presentation and this final journal.

Skills Developed

- Designing and verifying VLANs, inter-VLAN routing, ACLs, OSPF, and HSRP in Cisco Packet Tracer
- Translating an RFP into network requirements and an implementation plan
- Applying HIPAA and NIST guidance to a healthcare network design
- Estimating and validating project costs
- Presenting technical material to a client audience and working as part of a distributed team

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